Stochastic Calculus Home Assignment, School of Quants
November 9, 2025

- Throughout the text, let $W(t)$ represent Brownian motion, where $t \geq 0$.
- If you use Ito's lemma, write it out and identify each derivative explicitly.
- Combine your work into a single PDF file, including all relevant code.
- The assignment is lengthy, but each task is designed to deepen your understanding of a specific concept and guide you through the standard learning process:

  "I think I understand" → "Actually, I don't" → "Now I understand".
- It's better to solve something than nothing. Complete what you can to the best of your ability, and do not plagiarize
- If you have any question, contact me on Telegram: **@TimurShakurov**.
- Good luck!

1 Understanding Brownian motion and Ito's Integral

1. (5 points) **Warm up – scale on sale**

   Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\{W_t : t \geq 0\}$ be a standard Wiener process. Show that under normal scaling,

   $$B_t = cW_{t/c^2}, \quad c \neq 0$$

   is also a standard Wiener process.

2. (10 points) **Play with the clay**

   Let $I(t) = \int_0^t \mu(s) dW(s)$, where

   $$\mu(t) = \begin{cases} 1 & \text{if } t \in [0, 1); \\ 2 & \text{if } t \in [1, 2); \\ 5 & \text{if } t \in [2, 3); \\ -1 & \text{if } t \in [3, 4); \\ -5 & \text{if } t \in [4, 5). \end{cases}$$

   (a) (3 points) Simulate and plot one path of $W(t)$ and the corresponding path of $I(t)$. Plot $\mu(t)$ and mark the periods using dashed vertical lines.

   (b) (3 points) Simulate 300 paths of $W(t)$ and corresponding $I(t)$. Using two different colors for $W(t)$ and $I(t)$, plot both processes on another graph below (make the lines semi-transparent so the distribuiton is visible). Plot $\mu(t)$ and display periods with dashed vertical lines.

   (c) (4 points) Plot sample and analytical mean, 5% and 95% quantiles. Compare them. For the mean, and then for the quantiles, choose a sufficient number of simulations, so that sample mean/quantiles match nicely with the analytical mean/quantiles. Comparing visually is enough, no need to use strict statistical criteria, but feel free to use one if you want to.

We usually understand things better when we can touch them and play with them. Now you understand, exactly, what Itô's integral for a simple integrand really is.

3. (25 points) **Don't be so mean**
For $t \geq 0$, let $W(t)$ be a Brownian motion and $B(t) = W(t) - \int_0^t W(s)ds$.
   (a) (5 points) Express $B(t)$ as an Ito integral (look at the definition of the Ito ~~process~~ integral).
   Comment: $B(t)$ can be written as an Itô process: $B(t) = W(t) - \int_0^t W(s)ds = \int_0^t (-W(s))\, ds + \int_0^t 1dW(s)$. The task, however, is to express $B(t)$ purely as an Ito integral.
   Hint: represent both $W(t)$ and $\int_0^t W(s)ds$ as Ito's integrals. For the latter, apply Ito's lemma to $tW(t)$.
   (b) (5 points) Find the distribution of $B(t)$.
   (c) (5 points) For $t = 1$, examine the expression from part (a) along with its discretized version. Provide an intuitive explanation for these random variables. Are there any real-life random variables that you believe are analogous to the discretized or continuous version of $B(1)$?
   (d) (5 points) Simulate one path of $W(t)$ and the corresponding path of $B(t)$. Plot both paths.
   (e) (5 points) Using two different colors for $W(t)$ and $B(t)$, plot both processes on a separate graph below (make the lines semi-transparent so the distribution is visible).
   (f) (0 points) *This task is not to be graded but you will get feedback.*
   Brain teaser: can you express tW(t) as an Ito process? Can you express $tW(t)$ as an Ito integral? Why do they differ? Is $tW(t)$ a martingale? Verify by taking the conditional expectation. Are all Ito integrals – martingales, and if not, what is the catch?

The hope is that now you have strong intuition for Ito's Integrals.

2 Applying Ito's Lemma and solving practical models

4. (20 points) **Life brings ~~di~~lemmas**
Let $dS(t) = \mu(t)S(t)dt + \sigma(t)S(t)dW(t)$, where $\mu(t)$ and $\sigma(t)$ – some random processes.
   (a) (5 points) Find an explicit solution for $S(t)$.
   Hint: apply Ito's lemma to $ln(S(t))$.
   (b) (5 points) What can you say about the distribution of $S(t)$?
   (c) (5 points) Make a simplifying assumption (if needed) and find the expectation and the variance of $S(t)$.
   (d) (5 points) Calculate the median. Explain its behavior assymptotically. Compare to the mean.

5. (15 points) **Put t in theta**

   Consider the Vasicek model. Assume that $\theta = \theta(t)$ is a time-varying (possibly random) process.

   (a) (5 points) Derive the explicit solution for $r(t)$.

   Hint: look at the solution of the standard Vasicek model (constant θ).

   (b) (5 points) Calculate the expectation and the variance of the process.

   (c) (5 points) Describe the obvious cases when the process is Normally distributed (conditional on $r(0)$).

   In later courses you will learn why it's important for the $\theta(t)$ parameter to be time-varying. Spoiler – it allows fitting the initial yield curve without arbitrage.

3 Girsanov's theorem

6. (25 points) **Arithmetic Brownian Motion.**

   Consider an economy consisting of a risk-free asset and a stock price (risky asset). At time t, the risk-free asset B_t and the stock price S_t have the following dynamics:

$$dB_t = r_t B_t \, dt, \qquad dS_t = \mu_t \, dt + \sigma_t \, dW_t$$

   where r_t is the risk-free rate, μ_t is the stock price drift rate, and σ_t is the stock price volatility (which are all time dependent), and $\{W_t : 0 \leq t \leq T\}$ is a $\mathbb{P}$-Brownian Motion. on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

   From the discounted stock price process $X_t = e^{-\int_0^t r_u \, du} S_t$

   (a) (15 points) Find the risk-neutral measure (show how exactly you get this measure and which theorems you use).

   (b) (10 points) For the processes $W(t), \tilde{W}(t), X(t), S(t)$, write down (make a 4x4 table):

   - both differential and integral forms
   - expressed via both $\tilde{W}(t)$ and $W(t)$

   Specify which process is a martingale under which measure. Explain every transition and theorem you use.

   Hint: to get $X(t)$, apply Ito's product rule.